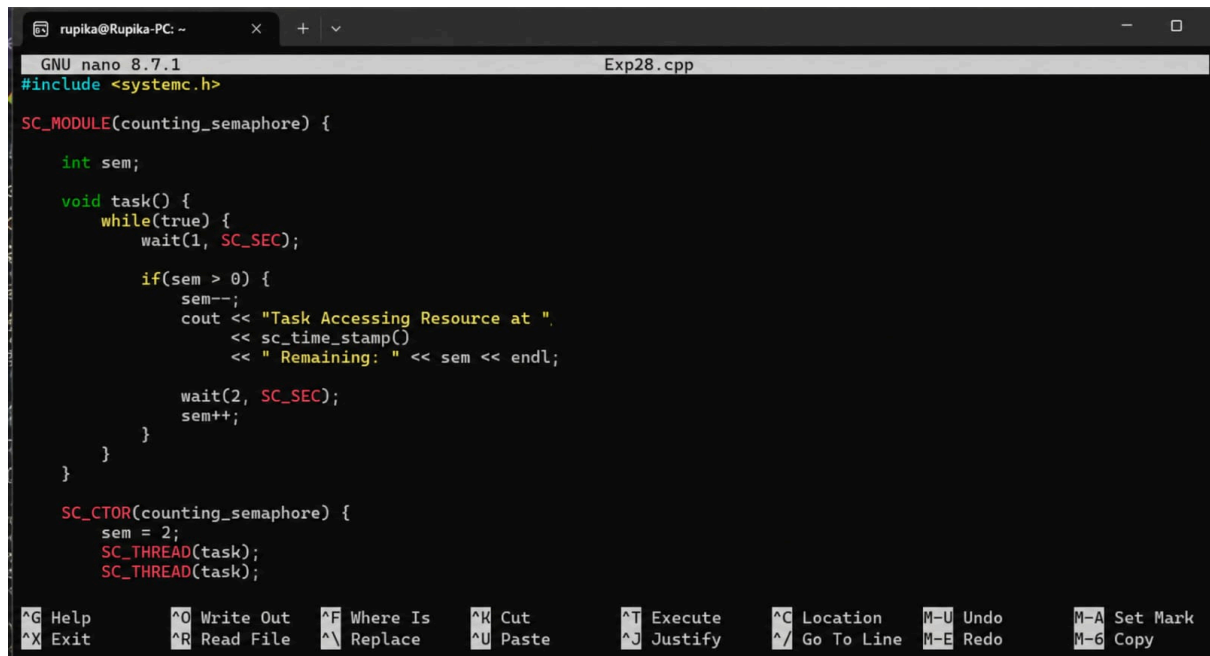


EXP:28



```
rupika@Rupika-PC: ~  
GNU nano 8.7.1 Exp28.cpp  
#include <systemc.h>  
  
SC_MODULE(counting_semaphore) {  
  
    int sem;  
  
    void task() {  
        while(true) {  
            wait(1, SC_SEC);  
  
            if(sem > 0) {  
                sem--;  
                cout << "Task Accessing Resource at "  
                    << sc_time_stamp()  
                    << " Remaining: " << sem << endl;  
  
                wait(2, SC_SEC);  
                sem++;  
            }  
        }  
    }  
  
    SC_CTOR(counting_semaphore) {  
        sem = 2;  
        SC_THREAD(task);  
        SC_THREAD(task);  
    }  
}
```

Help **Write Out** **Where Is** **Cut** **Execute** **Location** **Undo** **Set Mark**
Exit **Read File** **Replace** **Paste** **Justify** **Go To Line** **Redo** **Copy**

```
rupika@rupika-PC:~$ nano Exp28.cpp
rupika@rupika-PC:~$ g++ -o Exp28 Exp28.cpp -lsystemc
rupika@rupika-PC:~$ ./Exp28
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
```

```
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: kernel/sc_object_manager.cpp:153
```

```
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: kernel/sc_object_manager.cpp:153
```

```
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
```

```
rupika@rupika-PC:~$ |
```